IV. Test of Hypothesis

Statistical Hypothesis :-

→ We know that the to find unknown population parameter is called Estimation but To decide about population parameter on the basis of simple statisitics. (sample information), If make certain assumptions about the population.

→ These assumptions which true or maynot be true.

→ To Estimate the value of a parameter we should test a Hypothesis about a parameter.

→ The Hypothesis being a reffered as 'H'.

Null Hypothesis :- (H_0)

The Null Hypothesis H_0 is a statistical statement that assumes, there is no significant difference or effect b/n The variables ~~mean~~ being study.

Ex:-1) H_0 :- there is no significance reference in student's performance in method A & method B.

2) The new medicine as no significant effect compare with Existing treatment.

Alternative Hypothesis :- (H_1)

- The Alternate Hypothesis (H_1) is a statistical Statement that suggests there is a significant reference (or) Effective b/n The variables being the studeied.

Ex:-) H_0 :- There is no significance reference b/n method A and method B.

H_1 : There is a significance difference b/n method A & B.

2) H_0 : The new medicine has no significant effect.

H_1 : The new medicine has a significant effect.

Type 1 Error :

If the hypothesis it is true, But it is rejected in error and is called type 1 Error.

(or)

we can say rejection of the null hypothesis when it is true it is called a type 1 - Error.

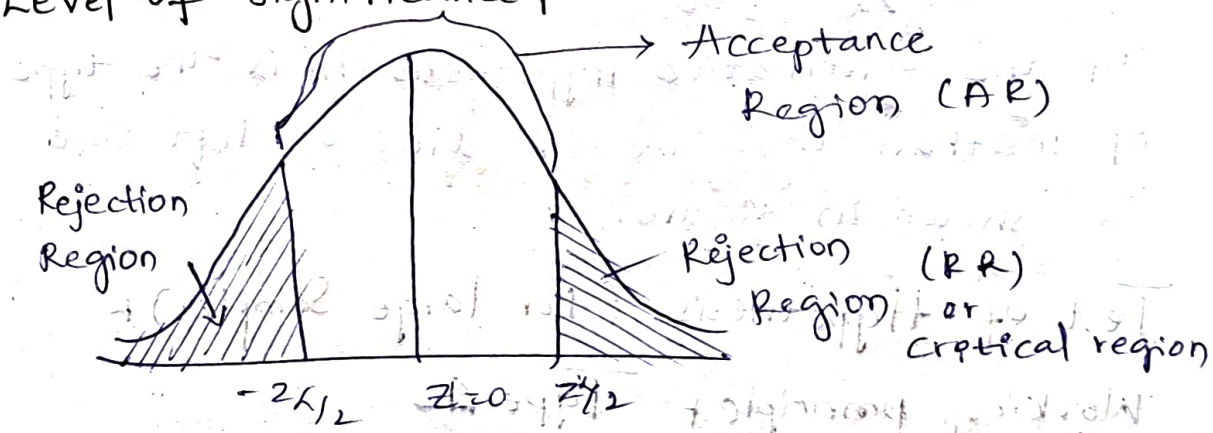
Type 2 - Error :

If the hypothesis is false, But is accepted in error and it is called Type 2 - Error.

(or)

we can say acceptance of the null hypothesis when it is false it is called Type 2 - Error.

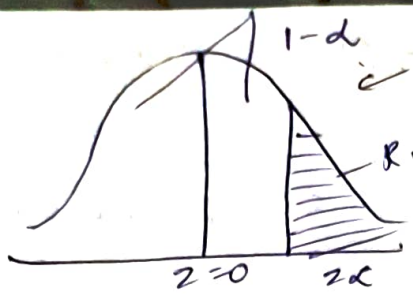
Level of Significance :



⇒ The probability of value of the variate falling in the critical region is the level of significance.

Critical region / Rejection region :

⇒ The critical region is that in which a sample value calling is rejecting



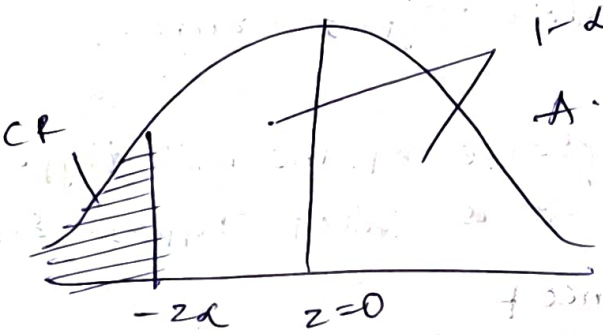
⇒ Right-tailed test

If the Alternative hypothesis H_1 is type of not equal then Critical region lies on both sizes. (Right & left)

⇒ Two-tailed Test (RTT)

If the alternative hypothesis H_1 is the type of greather than the critical region lies on the right side, as shown in figure.

⇒ Left-tailed Test (LTT)



If the Alternative Hypothesis H_1 is the type of lessthan than the CR lies on left side as shown in figure.

Test of Hypothesis (For large Samples)

Working principle: Type 1

1) Single population mean μ, with no variance σ^2.

For the large sample ($n \geq 30$), to test whether the population mean equals to a constant μ_0 or not.

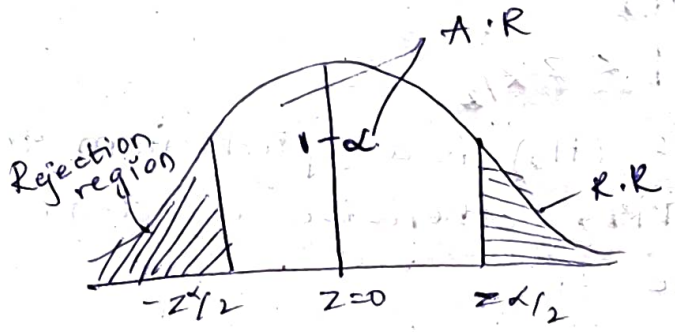
step 1 : Null Hypothesis (H_0) : $\mu = \mu_0$

" 2. Alternative Hypothesis (H_1) : $\mu \neq \mu_0$ (or) $\mu > \mu_0$ (or) $\mu < \mu_0$

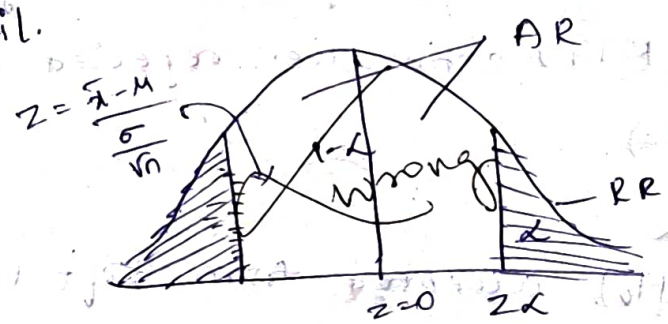
" 3. Level of significance : α (or) $1-\alpha$

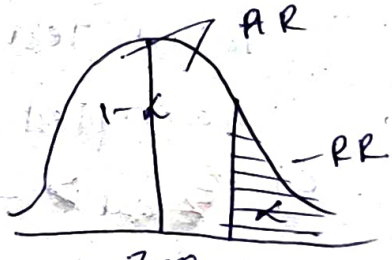
" 4. Critical region :-

a) If $\mu \neq \mu_0$ Test is two tailed Test for given α, critical values are $-z_{\alpha/2}$ and $z_{\alpha/2}$ from normal distribution tail.

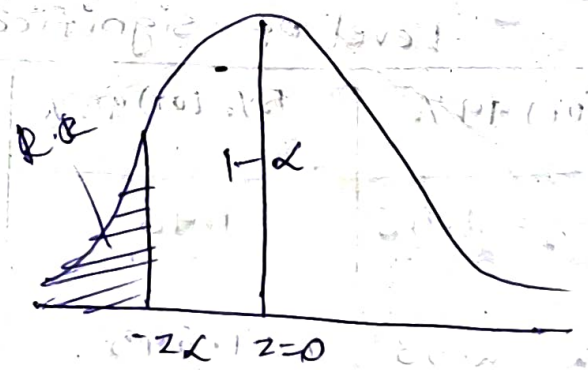


b) If $\mu > \mu_0$, test is right tailed test for given α, Critical value is z_α from the normal distribution tail.





c) If $\mu < \mu_0$ test is left-tailed test, the critical value is $-z_\alpha$ from the normal distribution tail.



5. Test of statstic :-

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

where; $\bar{x}$ = mean of the sample
μ = population mean
σ = population S.D
n = sample size.

6. Decision (or) conclusion :-

a) If $-z_{\alpha/2} < z_{cal} < z_{\alpha/2}$

(or)

$|z_{cal}| < z_{\alpha/2}$

Null Hypothesis (H_0) is accepted from two tailed Test (TTT) otherwise rejected. (accepted the H_1)

b) $z_\alpha > z_{cal}$

Null Hypothesis H_0 is accepted from Right tailed Test (RTT) otherwise rejected (i,e accepted H_1)

c) $z_{cal} > -z_\alpha$

Null hypo. (H_0) accepted from Left tailed Test (LTT) otherwise rejected (i,e accepted H_1)

Critical Values :-

Basis.	Level of significance.		
	1% (or) 99%	5% (or) 95%	10% (or) 90%
Critical values for two tailed test	$z_{\alpha/2} = 2.58$	1.96	1.645
Critical values for right tailed test	$z_\alpha = 2.33$	1.645	1.28
Critical values for left tailed test	$-z_\alpha = -2.33$	-1.645	-1.28

1) A sam
and S.D
from a
SD 2.61
its mea
limits

sol:- G

step 1:-
Sample

step 2:-
not been

Step 3:-

step 4

step

1) A sample of 900 members has a mean of 3.4 cms and S.D 2.61 cm. Is this sample has been taken from a large population of mean 3.25 cms. and SD 2.61 cms. If the population is normal and its mean is unknown, find the 95% fiducial limits of true mean.

Sol: Given that $n = 900$, $\bar{x} = 3.4$, $S = 2.61$ | $\mu = 3.25$, $\sigma = 2.61$

Step 1: Null hypothesis (H_0): Assume that the sample has been from the population mean $\mu = 3.25$

i.e $H_0 : \mu = 3.25$

Step 2: Alternative hypo (H_1): sample that has not been from population mean $\mu \neq 3.25$

i.e $H_1 : \mu \neq 3.25$ (two tailed test) of Hypo.

Step 3: Level of significance:

$\alpha = 5\%$.

(or) $1 - \alpha = 95\%$.

$\Rightarrow Z_{\alpha/2} = 1.96$.

Step 4: Test statistics:

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \Rightarrow Z = \frac{3.4 - 3.25}{\frac{2.61}{\sqrt{900}}}$$

$Z = 1.7241$

$Z_{cal} = |Z| = 1.7241$ (two tailed test)

Step 5: Conclusion:

$Z_{\alpha/2} = 1.96$

$Z_{cal} = |Z| = 1.7241$

$\therefore |Z| < Z_{\alpha/2}$

$\Rightarrow 1.7241 < 1.96$.

$\therefore$ we accept the null hypothesis H_0.

i.e the sample has been drawn from the population with mean $\mu = 3.25$

$\Rightarrow$ 95% confidence limits for the mean is given by

$$= \left[\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

$$= \left[3.4 - 1.96 \times \frac{2.61}{\sqrt{900}}, 3.4 + 1.96 \times \frac{2.61}{\sqrt{900}}\right]$$

$$= [3.2295, 3.57]$$

2) According to the norms established mechanical Aptitude test persons who are 18 years old have an average Height of 73.2 with a standard deviation of 8.6. If 4 randomly selected persons of that age averaged 76.7 test the Hypothesis $\mu = 73.2$ against The Alternative hypo. $\mu > 73.2$ at the 0.01 level of significance

Sol: Given that $n = 4$ | $\mu = 73.2$

$\bar{x} = 76.7$ | $SD = 8.6$

Step 1: Null Hypothesis : $\mu = 73.2$

Step 2: Alternative Hypothesis (H_1): $\mu > 73.2$

(Right-tailed test)

Step 3: level of significance:

$\alpha = 0.01$ (or) 1%.

$1 - \alpha = 0.99$ (or) 99%.

$\Rightarrow z_\alpha = 2.33$

Step 4: Test of statistic: $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$$\Rightarrow z = \frac{76.7 - 73.2}{\frac{8.6}{\sqrt{4}}} = 0.8139$$

$Z_{cal} = Z = 0.8139$

Step 5: conclusion :

$Z_{cal} = Z = 0.8139$

$Z_\alpha = 2.33$

$\therefore Z_{cal} < Z_\alpha$

$0.8139 < 2.33$ ✓

Type 2 : It always two tailed test (not depends on question).

⇒ Test of Hypothesis for difference of means of two large samples.

let $\bar{x}_1$ and $\bar{x}_2$ be the sample means of two independent large random samples.

sizes n_1 and n_2 drawn from pop? having means μ_1 and μ_2 ; SD's σ_1 and σ_2

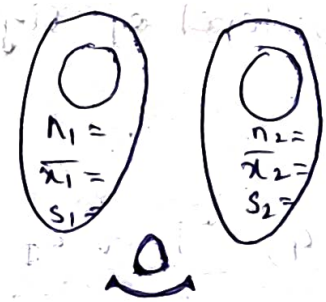
⇒ To test weather two population means are equal.

1) $H_0 \Rightarrow \mu_1 = \mu_2$ (or) $\bar{x}_1 = \bar{x}_2$

2) $H_1 \Rightarrow \mu_1 \neq \mu_2$ (or) $\bar{x}_1 \neq \bar{x}_2$

3) Test of statistic :

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$



a) S.E of difference of means.

$$S.E\ (\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

b) $$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

⇒ SD's (2 standard deviations are given in question) s_1 & s_2

c) $\sigma_1^2 = \sigma_2^2 = \sigma^2$ → equal variance

where $\sigma^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}$

$$\therefore Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

1) Sample of students were drawn from 2 universities and from their weights in kg's, mean & S.D are calculated and shown below, Make a large sample to test the significance of difference b/n the means.

	Mean	SD	Size of the Sample
University-A	55	10	400
University-B	57	15	100

Sol:- Given that $\bar{x}_1 = 55$; $\bar{x}_2 = 57$

$S_1 = 10$; $S_2 = 15$

$n_1 = 400$; $n_2 = 100$

1) Null hypothesis (H_0) : $\mu_1 = \mu_2$ (or) $\bar{x}_1 = \bar{x}_2$

2) Alternative hypo (H_1) : $\bar{x}_1 \neq \bar{x}_2$ ($\mu_1 \neq \mu_2$)

3) level of significance : $\alpha = 5\%$

(or) $1 - \alpha = 95\%$.

$Z_{\alpha/2} = 1.96$

4) Test of statistics :

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{(n_1)} + \frac{S_2^2}{(n_2)}}}$$

$$Z = \frac{55 - 57}{\sqrt{\frac{10^2}{400} + \frac{15^2}{100}}} = -1.26$$

$Z_{cal} = |Z| = 1.26$.

5. Conclusion :- $Z_{cal} = |Z| = 1.26$

$Z_{\alpha/2} = 1.96$

$\therefore |Z| < Z_{\alpha/2}$

$1.26 < 1.96$

$\therefore$ we accept the null hypothesis H_0.

There is no significance, b/n two means.
difference.

2) The means of two large samples of sizes 1000 and 2000 members are 67.5 Inches, and 68.0 Inches respectively. can the samples be recorded from the same population for S.D 2.5 Inches.

Sol:- Given that $n_1 = 1000$ | $n_2 = 2000$

$\bar{x}_1 = 67.5$ | $\bar{x}_2 = 68.$

$\sigma = 2.5$

Step 1:- Null hypothesis :- (H_0) :- $\mu_1 = \mu_2$ (or) $\bar{x}_1 = \bar{x}_2$

Step 2:- Alternative Hypo (H_1) :- $\bar{x}_1 \neq \bar{x}_2$ (Two tailed test)

Step 3:- level of significance :-

$\alpha = 5\%$

$1 - \alpha = 95\%$.

$Z_{\alpha/2} = 1.96$.

Step 4:- Test Statistics :-

$$Z = \left(\frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \right)$$

$$Z = \frac{67.5 - 68}{2.5 \sqrt{\frac{1}{1000} + \frac{1}{2000}}} \Rightarrow \frac{-0.5}{2.5 \sqrt{\frac{3}{2000}}}$$

$$\frac{-0.5}{2.5 \times 0.0387} = \frac{-0.5}{0.09675}$$

$Z = -5.167$

$|Z| = 5.16$

5) conclusion :-

$Z_{cal} = |Z| = 5.16$

$Z_{\alpha/2} = 1.96$

$\therefore |Z| < Z_{\alpha/2}$

$5.16 < 1.96$

∴ we reject the null hypo. (H_0) and accept the alternative hypo (H_1)

∴ There is significance difference b/n the 2 sample means.

Type-3

Test of Significance for single Proportion – Large samples

Suppose a large random sample of size 'n' has a sample proportion 'p' of nos Pocessing a certain attribute

→ Proportion of success

To test the hypothesis, that the Proportion 'P' in the population has a specified value 'P_0'

Null hypothesis $H_0 : P = P_0$

Alternate hypothesis $H_1 = P \neq P_0, P > P_0, P < P_0$

Test statistics:-

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} \quad \text{where } p = x/n$$

Note:-

1) $S.E(P) = \sqrt{\frac{PQ}{n}}$

2) confidence interval for True Proportion = $\left[P - Z_{\alpha/2}\sqrt{\frac{PQ}{n}}, P + Z_{\alpha/2}\sqrt{\frac{PQ}{n}}\right]$

→ Among 900 Peo
to be chap
true Proporti
Given that
Level o
Confide
2) In a sa
and the test
what are
Give
540
P =
small
Big
step

.16

96

$Z_{\alpha/2}$

1.96

null hypo. (H_0) and accept the

ypo (H_1)

icance difference b/n the 2 Sample means.

Proportion – Large samples

ample of size 'n' has a sample proportion

tain attribute

Proportion 'P' in the population has a

, $P > P_0$, $P < P_0$

ue Proportion = $\left[P - Z_{\alpha/2}\sqrt{\frac{PQ}{n}}, P + Z_{\alpha/2}\sqrt{\frac{PQ}{n}}\right]$

→ Among 900 People (Among 900 Peoples) from a state 90 are found to be chapathi eaters. Construct 99% confidence interval for the true Proportion.

Given that $n = 900$

$x = 90$

$$P = x/n = \frac{90}{900} = \frac{1}{10} = 0.1$$

Level of Confidence $1 - \alpha = 99\% \Rightarrow \alpha = 1\%$

$$Z_{\alpha/2} = 2.58$$

Confidence interval for true Proportion

$$= \left[P - Z_{\alpha/2}\sqrt{\frac{PQ}{n}}, P + Z_{\alpha/2}\sqrt{\frac{PQ}{n}}\right]$$

$$= \left[0.1 - 2.58\sqrt{\frac{(0.1)(0.9)}{900}}, 0.1 + 2.58\sqrt{\frac{(0.1)(0.9)}{900}}\right]$$

$$= [0.0742, 0.1258]$$

2) In a sample of 1000 people in karnataka 540 are rice eaters and the rest all wheat eaters. Can we assume that both rice and what are equally popular in these state 1% level of Significance.

Given $n = 1000$

540 rice eaters out of 1000

Small p = Sample proportion of rice eaters $= \frac{540}{1000} = 0.54$

Big P = (let assume that equally eaters)

population proportion of rice eaters $= \frac{P=2}{} = 1/2 = 0.5 = P_0$

step 1: H_0 = Both Rice & wheat are equally Popular in the state

$H_0 = P = 0.5$

Step 2:- $H_1 : P \neq 0.5$ (two tailed test)

Step 3:- Level of significance : $1 - \alpha = 99\%$.

$$\alpha = 1\%.$$

$$\Rightarrow Z_{\alpha/2} = 2.58$$

Step 4:- Test Statistic

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.54 - 0.5}{\sqrt{\frac{(0.5)(0.5)}{1000}}}$$

$$Z = 2.532$$

$$\therefore Z_{cal} = |Z| = 2.532$$

Step 5:- Conclusion:-

$Z_{cal} = |Z| = 2.532$

$Z_{tab} = Z_{\alpha/2} = 2.58$

$|Z| \leq Z_{\alpha/2}$

(or)

$-Z_{\alpha/2} < Z < Z_{\alpha/2}$

$= 2.532 < 2.58$

It accepts the null hypothesis H_0 at 1% level of significance

Rice and wheat are equally popular in karnataka state

3) In a big city 325 men out of 600 men were found to be smokers. Thus, this information supports the conclusion that the majority of men in the city are smokers.

Given x = no. of smokers $s = 325$, $n = 600$

$$p = \frac{x}{n} = \frac{325}{600} = 0.5417$$

P = population proportion of smokers in the city $= 1/2$

$= 0.5$

$Q = 0.5$ (non-smokers)

Thus this information supports majority

Step 1: Null Hypothesis

$H_0 = P = 1/2$ (smokers)

The no. of smokers equals with no. of non-smokers

Step 2) $H_1 = P > 1/2$ (R.T.T)

3) Level of Significance:-

$1 - \alpha = 95\%$

$\alpha = 5\%$

$Z_\alpha = 1.96 \quad 1.645$

$$\boxed{Z_\alpha = 1.645}$$

4) Test statistics:-

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.5417 - (0.5)}{\sqrt{\frac{(0.5)(0.5)}{600}}} = 2.04$$

$Z_{cal} = 2.04$ (R.T.T)

(5) Conclusion

$Z_{cal} = Z = 2.04$

$Z_{table} \equiv Z_\alpha = 1.645$

$Z_{cal} < Z_{tab}$ i.e $Z < Z_\alpha$

$2.04 < 1.645$ (✗) not-satisfied

Rejects null hypothesis, Accepts alternative hypothesis H_1

That is the majority of men in city are smokers.

Type-4:-

Test for equality of 2 proportions for Large Samples.



1) $H_0 : P_1 = P_2$

2) $H_1 : P_1 \neq P_2$

3) Test statistic:-

$$Z = \frac{P_1 - P_2}{\sqrt{\frac{P_1 q_1}{n_1} + \frac{P_2 q_2}{n_2}}} \quad (\text{or}) \quad Z = \frac{P_1 - P_2}{\sqrt{\frac{P_1 q_1}{n_1} + \frac{P_2 q_2}{n_2}}} \quad (\text{or})$$

$$Z = \frac{P_1 - P_2}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \quad \text{where } p = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2} \quad q = 1 - p$$

Random Samples of 400 men and 600 women were asked whether they would liked to have flyover near their residence. 200 men and 325 women were in favour of proposal. Test the hypothesis, that proportion of men & women in favour of proposal are equal where 5% level of Signif

$n_1 = 400$ | $n_2 = 600$

$x_1 =$ | $x_2 =$

$P_1 = \frac{x_1}{n_1} = \frac{200}{400}$ (men) | $P_2 = \frac{x_2}{n_2} = \frac{325}{600}$ (women)

$P_1 \rightarrow$ proportion of men

$P_2 \rightarrow$ " " " women

Step 1) Null Hypothesis

$H_0 : P_1 = P_2$ (or $P_1 - P_2$)

i.e that is assum that there is no significant difference b/w the option of men and women as far as proposal of flyovers

H_1:- Alternative hypothesis

$H_1 : P_1 \neq P_2$

Step 3:- Level of significance (T.T.T) $\Rightarrow 1-\alpha = 95\%$,

$\alpha = 5\% = 1.96$

Step 4:- Test statistics:-

$$P = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2} = 0.525$$

$$Z_{cal} = \frac{P_1 - P_2}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = |Z| = 1.3$$

$q = 1 - p = 1 - 0.525 = 0.475$

$$Z = \frac{0.5 - 0.5416}{\sqrt{(0.475)(0.525)\left(\frac{1}{400} + \frac{1}{600}\right)}} = -1.3$$

$|Z| = 1.3$ (T.T.T)

Conclusion:-

$Z_{cal} = |Z| = 1.3$

$Z_{tab} = Z_{\alpha/2} = 1.96$

$\therefore |Z| < Z_{\alpha/2}$

$1.3 < 1.96$

It accepts the null hypothesis.

men & women are equally supporting.

2) On the basis of their total scores 200 candidates of a civil Service Examination are divided into 2 groups, the upper 30 Percentage and remaining 70%. Consider the 1st question of Examination. Among the first group, 40 had correct answers whereas among 2nd group, 80 had correct answer. On the basis of these results, can one conclude that 1st question is not good at discriminative ability of type being examined Here.

Sol:- we have $n_1 = 60$; $n_2 = 140$

$x_1 = 40$; $x_2 = 80$ and

$P_1 = \frac{x_1}{n_1} = \frac{40}{60} = \frac{2}{3} = 0.667$

$P_2 = \frac{x_2}{n_2} = \frac{80}{140} = 0.571$

$\therefore p = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{40+80}{60+140} = \frac{120}{200}$

$p = 0.6$

and $q = 1 - p \Rightarrow 1 - 0.6$

$= 0.4$

1) Null Hypothesis $H_0 \Rightarrow P_1 = P_2$

2) Alternative hypothesis $H_1 \Rightarrow P_1 \neq P_2$

3) Level of significance $\alpha = 0.05 \Rightarrow 5\%$ (or) 95%

4) Test statistic is

$$Z = \frac{P_1 - P_2}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$\Rightarrow \frac{0.667 - 0.571}{\sqrt{(0.6)(0.4)\left(\frac{1}{600} + \frac{1}{400}\right)}} = \frac{0.096}{0.1} = 0.96$$

5) Conclusion:

$$Z_{cal} = |Z| = 0.96$$

$$Z_{tab} = Z_{\alpha/2} = 1.96$$

$$\therefore |Z| < Z_{\alpha/2}$$

$$0.96 < 1.96$$

It accepts the Null Hypothesis (H_0).

Test of Hypothesis - for Small Samples.

⇒ Type 5 :- (student's test for single mean - Small Sample).

Test statistic :-

$$t = \frac{\bar{x} - \mu}{\frac{S}{\sqrt{n-1}}} \quad (\text{or}) \quad t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

where 's' is the sample SD follows t-distribution with $\nu = n-1$ degrees of freedom

1 Q) A sample 26 bulbs gives a mean life of 990 hours with S.D of 20 hours. The manufacturer claims that the mean life of bulbs is 1000 hrs. Is the sample not upto the standard

Sol:- Given $n = 26$

$\bar{x} = 990$

$S = 20$

$\mu = 1000$

Step 1 :- Null Hypothesis $H_0 \Rightarrow \mu = 1000$.

Step 2 :- Alternative hypo. $(H_1) \Rightarrow \mu \neq 1000$ (two-tailed test)

Step 3 :- Level of Significance :-

$1 - \alpha = 95\% \Rightarrow \alpha = 5\% = 0.05$

$t_{\alpha/2} = t_{0.025} \quad \alpha/2 = 0.025$

$= 26 - 1$

$\nu = 25$

$t_{\alpha/2} = t_{0.025} = 2.06$.

Step 4 :- Test statistic

$$t = \frac{\bar{x} - \mu}{\frac{S}{\sqrt{n-1}}} \Rightarrow \frac{990 - 1000}{\frac{20}{\sqrt{26-1}}}$$

$t = -2.5 \Rightarrow |t| = 2.5$

$t_{cal} = |t| = 2.5$

5. conclusion :-

$t_{cal} = |t| = 2.5$

$t_{tab} = t_{\alpha/2} = 2.06$

$|t| < t_{\alpha/2}$

i.e $t_{cal} < t_{tab}$.

$\Rightarrow 2.5 \not< 2.06$

Reject the null Hypothesis H_0 and accept the alternative hyp. (H_1).

2) The average breaking strength of the steel rod is specified to be 18.5 thousand pounds. to test this sample of 14 rods were tested. The mean and SD obtained were 17.85 and 1.955 resp". Is the result of exp. significant.

Sol:- Given $n = 14$; $\bar{x} = 17.85$

$S = 1.955$

$\bar{x}$ = sample mean

sample SD = S

n = sample size

pop. mean $\mu = 18.5$ and

degress of fredom (d.o.f) $\Rightarrow \nu = n - 1$

$= 14 - 1 = 13$

Step 1 :- Null Hypothesis $(H_0) = \mu = 18.5$

Step 2 :- Alternative Hypothesis $(H_1) = \mu \neq 18.5$

(two-tailed test)

Step 3 :- Level of significance :-

$1 - \alpha = 95\%$

$\alpha = 5\% = 0.05$

$\alpha/2 = 0.025$

$t_{\alpha/2} = t_{0.025} = 2.160$

$\nu = 13$

Step 4:- Test statistic

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{17.85 - 18.5}{\frac{1.955}{\sqrt{13}}}$$

$$t = -1.199$$

$$|t| = 1.199$$

$$t_{cal} = |t| = 1.199$$

Step 5:- Conclusion :-

$$t_{cal} = |t| = 1.199$$

$$t_{tab} = t_{\alpha/2} = 2.16$$

$$|t| < t_{\alpha/2}$$

$$t_{cal} < t_{tab}$$

$$1.199 < 2.16$$

∴ Accept the Null hypo. (H_0).

3) The heights of 10 mates of a given locality are found to be 70, 67, 62, 68, 61, 68, 70, 64, 64, 66 inches. Is It reasonable to belive that The avg. height is greather than 64 inches? Test at 5% significance level assuming that for '9' degrees of freedom ($t = 1.833$ at $\alpha = 0.05$)

Sol:- Mean $\bar{x} = \frac{\sum x}{n}$

$$= \frac{70 + 67 + 62 + \dots + 64 + 66}{10}$$

$$= \frac{660}{10} = 66$$

$$\boxed{\bar{x} = 66}$$

x	$x - \bar{x}$	$(x - \bar{x})^2$
70	$70 - 66 = 4$	16
67	$67 - 66 = 1$	1
62	-4	16
68	2	4
61	-5	25
68	2	4
70	4	16
64	-2	4
64	-2	4
66	0	0
$\sum x = 660$		90

we know that, $s^2 = \frac{1}{n-1} \sum (x - \bar{x})^2$

$$= \frac{1}{9} \times 90$$

$$s^2 = 10$$

$$\therefore s = \sqrt{10} = 3.16$$

$$\boxed{s = 3.16}$$

Step 1:- Null Hypothesis (H_0) $\Rightarrow \mu = 64$

Step 2:- Alternative Hypo (H_1) $\Rightarrow \mu > 64$

(Right tailed test)

Step 3:- level of significance :- $1 - \alpha = 95\%$

$$\alpha = 0.05$$

$$t_\alpha = t_{0.05} = 1.833$$

with $v = 10 - 1 = 9$

$v = 9$

Step 4:- Test statistic :-

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{66 - 64}{\frac{3.16}{\sqrt{10}}}$$

$$t = 2.02$$

$$t_{cal} \Rightarrow t = 2.02$$

Step5:- conclusion

$t_{cal} = t = 2.02$

$t_{tab} = t_\alpha = 1.833$

$\therefore t < t_\alpha$

$\therefore t_{cal} < t_{tab}$

$2.02 < 1.833$

$\therefore$ Accepts the Null Hypo. (H_0)

Type-6:- Test for difference of means. (small samples)

$$t = \frac{\bar{x} - \bar{y}}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where $s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$

(or)

$$s^2 = \frac{1}{n_1 + n_2 - 2}\left[\sum (x - \bar{x})^2 + \sum (y - \bar{y})^2\right]$$

NOTE 1:-

* Standard Error of difference of means (for small samples)

$$S.E(\bar{x} - \bar{y}) = s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

* Maximum Error $= t_{\alpha/2} \cdot S \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$

* Confidence limits for $(\bar{x}_1 - \bar{x}_2)$:-

$$= \left[(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2}\, s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \;(x_1 - x_2) + t_{\alpha/2} \cdot s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}\right]$$

***

1) The sample of two types of electrical light bulbs for tested for length of life and following data were obtained.

type-I	type-II
Sample no. $(n_1) = 8$	$n_2 = 7$
sample sd $(s_1) = 36$ hrs	$s_2 = 40$ hrs.
sample mean $(\bar{x}_1) = 1234$ hrs	$\bar{x}_2 = 1036$ hrs.

- Is the difference in the means sufficient to warranty that type I is superior to type II regarding length of life:

Sol:- Given that

$n_1 = 8$	$n_2 = 7$
$s_1 = 36$ hrs	$s_2 = 40$ hrs
$\bar{x}_1 = 1234$ hrs	$\bar{x}_2 = 1036$ hrs

Step 1:- Null hypothesis (H_0) :- $\bar{x}_1 = \bar{x}_2$ ($\mu_1 = \mu_2$)

Step 2:- Alternative hypo (H_1) :- $\bar{x}_1 \neq \bar{x}_2$ ($\mu_1 \neq \mu_2$) (Two tailed test)

Step 3:- Level of significance:-

$1 - \alpha = 95\%$

$\alpha = 5\% = 0.05$

$\alpha/2 = 0.025$

$v = (n_1 - 1) + (n_2 - 1)$

$v = n_1 + n_2 - 2$

$v = 8 + 7 - 2$

$v = 13$

$\therefore t_{\alpha/2} = t_{0.025} = 2.160$

Step 4:- Test statistic

$$t = \frac{\bar{x} - \bar{y}}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{where } s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

$$s^2 = \frac{8(36)^2 + 7 \times (40)^2}{8 + 7 - 2}$$

$s^2 = 1659.076925$

$s = 40.73$

$$\therefore t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}$$

$$= \frac{1234 - 1036}{40.73\sqrt{\frac{1}{8}+\frac{1}{7}}}$$

$$t = 9.3928$$

$$|t| = 9.3928$$

$$t_{cal} = |t| = 9.3928$$

Step 5 :- Conclusion :-

$$t_{cal} = |t| = 9.3928$$

$$t_{tab} = t_{\alpha/2} = t_{0.025} = 2.160$$

$$\therefore |t| < t_{\alpha/2}$$

$$\Rightarrow 9.3928 \not< 2.160$$

∴ Reject the Null hypo. and the Accepts the Alternative Hypo (H_1)

2) Two Horses A and B where tested according to the time (in seconds) to run a particular track to the following data :

Horse A	28	30	32	33	33	29	34
Horse B	29	30	30	24	27	29	----

Test wheather the two Horses have the same running capacity.

Soln $n_1 = 7$; $n_2 = 6$

If first complete samples means and standard deviation.

$\bar{x}$ is the mean of the first sample

$$\bar{x} = \frac{1}{7}(28+30+32+33+33+29+34)$$

$$\bar{x} = \frac{1}{7}(219) = 31.286$$

$\bar{y}$ = Mean of the second sample

$$= \frac{1}{6}[29+30+30+24+27+29]$$

$$\bar{y} = 28.16$$

x	$x-\bar{x}$	$(x-\bar{x})^2$	y	$y-\bar{y}$	$(y-\bar{y})^2$
28	-3.286	10.8	29	0.84	0.7056
30	-1.286	1.6538	30	1.84	3.3856
32	0.714	0.51	30	1.84	3.3856
33	1.714	2.94	24	-4.16	17.3056
33	1.714	2.944	27	-1.16	1.3456
29	-2.286	5.226	29	0.84	0.7056
34	2.714	7.366			
		$\Sigma(x-\bar{x})^2 =$ 31.4358			$\Sigma(y-\bar{y})^2 =$ 26.8336

$$s^2 = \frac{1}{n_1+n_2-2}\left[\Sigma(x-\bar{x})^2 + \Sigma(y-\bar{y})^2\right]$$

$$= \frac{1}{11}[31.4358 + 26.8336]$$

$$s^2 = 5.23$$

$$s = 2.3$$

Step 1 :- Null hypo (H_0) $\Rightarrow H_0 = M_1 = M_2$

Step 2 :- Alternative hypo (H_1) $= M_1 \neq M_2$ (Two tailed Test)

Step 3 :- level of significance :-

$1-\alpha = 95\%$.

$\alpha = 5\% = 0.05$

$\alpha/2 = 0.025$

$t_{\alpha/2} = 0.025$

$v = n_1 + n_2 - 2$

$\boxed{v = 11}$

$t_{\alpha/2} = 0.025$

$\boxed{t_{\alpha/2} = 2.20}$ → table value

Step 4:- Test statistic :-

$$t = \frac{\bar{x} - \bar{y}}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$t = \frac{31.286 - 28.16}{2.3\sqrt{\frac{1}{7} + \frac{1}{6}}}$$

$t = 2.443$

$\therefore |t| = 2.443$

$t_{cal} = |t| = 2.443$

Step5 :- Conclusion :-

$t_{cal} = |t| = 2.443$

$t_{tab} = t_{\alpha/2} = 2.20$

$\therefore |t| < t_{\alpha/2}$

$2.443 \not< 2.20$

∴ Reject The Null Hypo, accepts The H_1

3) To compare two times of bumper guards, 6 of each kind were mounted on a gaard and then the car was run into a concrete wall.
the following on the costs of repairs.

Bumper Guard-I	107	148	123	165	102	119
Bumper Guard-II	134	115	112	151	133	129

use the 0.01 error to test wheather differe. nce b/n the means of these two samples is significance

F-Distribution

Sampling distribution of the ratio of two sample variance.

* F-distribution can be used to test the equality of serval population means, comparing sample variances.
* under the assumptions (Hypothesis) that the 2 normal population have the same variances.

i,e $\sigma_1^2 = \sigma_2^2$

NOTE: $F = \frac{S_1^2}{S_2^2}$ where, $S_1^2 > S_2^2$, $\sigma_1^2 = \sigma_2^2$

(or)

$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$ where $S_1^2/\sigma_1^2 > S_2^2/\sigma_2^2$ and $\sigma_1^2 \neq \sigma_2^2$

* $F = \frac{\text{Greater variance}}{\text{smaller variance}}$.

1) For an F-distribution, Find

a) $F_{0.05}$ with $V_1 = 7$ and $V_2 = 15$

b) $F_{0.01}$ with $V_1 = 24$ and $V_2 = 19$

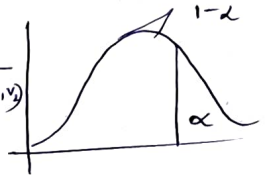
c) $F_{0.95}$ with $V_1 = 19$ and $V_2 = 24$

d) $F_{0.99}$ with $V_1 = 28$ and $V_2 = 12$

(a) Solr $F_{0.05}(V_1, V_2) = F_{0.05}(7, 15) = 2.71$

$F_{1-\alpha}(V_1, V_2) = \frac{1}{F_\alpha(V_1, V_2)}$

$= F_\alpha(V_2, V_1)$



(b) Solr $F_{0.01}(V_1, V_2) = F_{0.01}(24, 19) = 2.92$

(c) Solr $F_{0.95}(19, 24) = F_{1-0.05}(19, 24)$

$= \frac{1}{F_{0.05}(19, 24)} = F_{0.05}(24, 19)$ (1-α)

$= 2.11$

d) $F_{0.99}(28, 12) = F_{1-0.01}(28, 12)$

$= \frac{1}{F_{0.01}(28, 12)}$

$= F_{0.01}(12, 28) = 2.90$

$= 2.90_4$

2) Find the values of

a) $F_{0.05}$ for $V_1 = 15$ and $V_2 = 7$

b) $F_{0.95}$ for $V_1 = 12$ and $V_2 = 15$ (degrees of freedom)

c) $F_{0.95}$ for $V_1 = 10$ and $V_2 = 20$

d) $F_{0.99}$ for $V_1 = 6$ and $V_2 = 20$.

a) Solr $F_{0.05}(V_1, V_2) = F_{0.05}(15, 7) = 3.51$

$\neq \frac{1}{F_{0.05}(15, 7)}$

$\neq F_{0.05}(7, 15)$

(b) Solr $F_{0.95}(12, 15) = F_{1-0.05}(12, 15)$

$= \frac{1}{F_{0.05}(12, 15)} = F_{0.05}(15, 12)$

$= 2.62$

(c) Solr $F_{0.95}(10, 20) = F_{1-0.05}(10, 20)$

$= \frac{1}{F_{0.05}(10, 20)} = F_{0.05}(20, 10)$

$= 2.77$

(d) Solr $F_{0.99}(6, 20) = F_{1-0.01}(6, 20)$

$= \frac{1}{F_{0.01}(6, 20)} = F_{0.01}(20, 6)$

$= 7.40$

Properties of F-Distribution.

1) F-Distribution is free from population parameters and depends upon degrees of freedom (d.o.f) only.



2) F-distribution curve ~~entirely~~ lies entirely in 1st Qudrant.

3) The F-curve depends not only two parameters v_1 and v_2 but also on the order in which they are stated.

4) $F_{1-\alpha}(v_1, v_2) = \frac{1}{F_\alpha(v_1, v_2)}$

$= F_\alpha(v_2, v_1)$

Applications of F-distribution.

1) This is widely used in the Analysis of Variance (ANOVA)

2) This mainly used to test the hypothesis of equality between two variances.

3) F-test is also used to test the hypothesis of equality amongest different means.

Type-1 (F-test)

Test for equality of 2 population variances.

→ Let two Independent random samples of sizes, n_1 and n_2 be drawn from two normal populations.

→ To test the hypothesis that the two population variances σ_1^2 and σ_2^2 are equal.

Null: $H_0 \Rightarrow \sigma_1^2 = \sigma_2^2$ $(s_1^2 = s_2^2)$

Alter: $H_1 \Rightarrow \sigma_1^2 \neq \sigma_2^2$ $(s_1^2 \neq s_2^2)$

Test Statistic:

$$F = \frac{s_1^2}{s_2^2} \text{ where } s_1^2 > s_2^2$$

(or)

$$F = \frac{s_2^2}{s_1^2} \text{ where } s_2^2 > s_1^2$$

$$F = \frac{\text{Greater Variance}}{\text{Smaller Variance}}.$$

NOTE: When F is closed to 1, the two sample variances s_1 and s_2 are nearly same.

2) $F = \frac{s_1^2}{s_2^2}$ if $s_1^2 > s_2^2$ where $s_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{1}{n_1 - 1}$

$$= \frac{1}{n_1 - 1} \Sigma (x_i - \bar{x})^2 \leftarrow \Sigma (x_i - \bar{x})^2$$

$$s_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{1}{n_2 - 1} \Sigma (y_i - \bar{y})^2$$

***

1) The measurements of the output of two units have given the following Results. Assuming that both samples have been obtained from the normal population at 5% significance level, & Test weather 2 populations have the same results.

Unit A	14.1	10.1	14.7	13.7	14.0
Unit B	14.0	14.5	13.7	12.7	14.1

Sol: Step 1: Null hypo: (H_0): $\sigma_1^2 = \sigma_2^2$

Step 2: Alternative hypo $(H_1) = \sigma_1^2 \neq \sigma_2^2$

Step 3: level of significance $(1-\alpha)$ or $\alpha = 5\%$

$\alpha = 0.05$

$V_1 = 5-1 = 4$

$V_2 = 5-1 = 4$

$F_\alpha = F_{0.05}(4,4) = 6.39$

Step 4: Test statistic:

$$\bar{x} = \frac{14.1 + 10.1 + 14.7 + 13.7 + 14.0}{5}$$

$\bar{x} = 13.32$ and

$$\bar{y} = \frac{14.0 + 14.5 + 13.7 + 12.7 + 14.1}{5}$$

$\bar{y} = 13.8$

Unit - A

x	$x - \bar{x}$	$(x-\bar{x})^2$
14.1	$14.1 - 13.32 = 0.78$	0.6084
10.1	-3.22	10.3684
14.7	1.38	1.9044
13.7	0.38	0.1444
14.0	0.68	0.4624
		$\Sigma_{x_i} = 13.488$

Unit - B

y	$y - \bar{y}$	$(y-\bar{y})^2$
14.0	$14.0 - 13.8 = 0.2$	0.04
14.5	0.7	0.49
13.7	-0.1	-0.01
12.7	-1.1	-1.21
14.1	0.3	0.09
		$\Sigma(y-\bar{y})^2 = 1.84$

$$S_1^2 = \frac{1}{n_1 - 1} \Sigma(x_i - \bar{x})^2$$

$$= \frac{1}{4}(13.488)$$

$S_1^2 = 3.372$ and

$$S_2^2 = \frac{1}{n_2 - 1} \Sigma(y_i - \bar{y})^2$$

$$= \frac{1.84}{5-1} = 0.46 \quad S_2^2 = 0.46$$

Here $S_1^2 > S_2^2$

$$\therefore F = \frac{S_1^2}{S_2^2}$$

$$= \frac{3.372}{0.46}$$

$F = 7.33$

Step 5: conclusion

$F_{tab} = F_\alpha(V_1, V_2)$

$= F_{0.05}(4,4)$

$= 6.39$

and $F_{cal} = F = 7.33$

~~F_{tab}~~ $F_{cal} < F_{tab} \Rightarrow 7.33 \not< 6.39$

(or)

Rejected $F < F_\alpha$

~~Accepted~~ The Null hypo. (Rejected)

∴ There is significant difference b/n the variance

2) Pumpkins were under two Expermental condn of two random samples 11 and 9 pumpkins show the sample standard deviation of their weights are 0.8 and 0.5 resp", Assuming that the weight distribution are normal Test hypothesis that the two variance.

Sol:- Given that

$n_1 = 11$ | $n_2 = 9$

$S_1 = 0.8$ | $S_2 = 0.5$

step 1:- Null hypo $(H_0) \Rightarrow \sigma_1^2 = \sigma_2^2$

step 2:- Alternative hypo $(H_1) = \sigma_1^2 \neq \sigma_2^2$

step 3:- Level of significance :- $\alpha = 5\% = 0.05$

$V_1 = 11 - 1 = 10$ ~~= 3.35~~

$V_2 = 9 - 1 = 8 \Rightarrow F\alpha = F_{0.05}(10,8) = 3.35$

step 4:- Test statistic :-

$$S_1^2 = \frac{n_1 S_1^2}{n_1 - 1} = \frac{11 \times (0.8)^2}{11-1}$$

$S_1^2 = 0.704$ and

$$S_2^2 = \frac{n_2 S_2^2}{n_2 - 1} = \frac{9 \times (0.5)^2}{9-1}$$

$S_2^2 = 0.281$

$$F = \frac{\text{Greater variance}}{\text{Smaller Variance}} = \frac{S_1^2}{S_2^2}$$

$$F = \frac{0.704}{0.281} = 2.5$$

$\therefore F_{cal} = 2.5$

step 5:- conclusion

$F_{tab} = F\alpha\,(V_1, V_2)$

$= F_{0.05}\,(10,8)$

$= 3.35$

$F_{cal} = 2.5$

$F_{cal} < F_{tab} = 2.5 < 3.35$

∴ Accepts the Null hypo, Reject the Alter hypo

3) The Nicotine contains in milligram in a samples of tobacco were found to be as follows.

Sample A	24	27	26	21	25	—
Sample B	27	30	28	31	22	36

⇒ can it said that the 2 samples have come from the same normal population (or) same variance.

sol:- step 1 :- Null hypo (H_0) :- $\sigma_1^2 = \sigma_2^2$

step 2 :- Alternative hypo $(H_1) = \sigma_1^2 \neq \sigma_2^2$

step 3 :- level of significance $\alpha = 5\% = 0.05$

$V_1 = 5 - 1 = 4$

$V_2 = 6 - 1 = 5$

$F\alpha = F_{0.05}\,(4,5) = 5.19$

step 4 :- Test statistic :-

$\bar{x} = 24 + 27 + 26 + 21 + 25 = 123/5 = 24.6$

$\bar{y} = 27 + 30 + 28 + 31 + 22 + 36 = \frac{174}{6} = 29$

$\bar{x} = 24.6$ and $\bar{y} = 29$

Unit A:-

x	$x - \bar{x}$	$(x - \bar{x})^2$
24	−0.6	+0.36
27	2.4	5.76
26	1.4	1.96
21	−3.6	+12.96
25	0.4	+0.16

$\Sigma(x - \bar{x})^2 = +8.427$

Unit B :-

y	$y-\bar{y}$	$(y-\bar{y})^2$
29	-2	4
30	1	1
28	-1	1
31	2	4
22	-7	49
36	7	49

$\Sigma(y-\bar{y})^2 = 108$

$$s_1^2 = \frac{1}{n_1-1}\Sigma(x_i-\bar{x})^2$$

$$= \frac{1}{4}(8.427)$$

$\boxed{s_1^2 = 2.106}$ and

$$s_2^2 = \frac{1}{n_2-1}\Sigma(y_i-\bar{y})^2$$

$$= \frac{1}{5}(108) = 21.6$$

$\boxed{s_2^2 = 21.6}$

$$F = \frac{\text{Greater Variance}}{\text{Smaller Variance}} = \frac{s_1^2}{s_2^2} = \frac{2.106}{21.6}$$

$F = 0.0975$

$\boxed{F = 0.0975}$

Steps :- Conclusion

$f_{tab} = F_\alpha(v_1, v_2)$

$= F_{0.05}(4,5) = 5.19$

and $F_{cal} = F = 0.0975$

$F_{cal} < f_{tab} = 0.0975 < 5.19$

Accepts the Null Hypo, Reject the Alter Hyp

V. Applied Statistics

Curve fitting :-

Suppose a set of values (x_1, y_1) and (x_2, y_2) --- (x_n, y_n) are given where statisfy the equation.

$y_i = f(x_i)$ —① where $i = 1, 2, 3 \cdots n$

If these points are plotted, the dotted points are scattered. This diagram is called scattered diagrams.



Scattered digram.

→ To find the equation of curve $y = f(x)$ —② which statifies the eq ① is curve fitting.

Here, our Aim is to find the functional relationsh. 'ip b/n the observed values (eqn ①) and ~~excepted~~ expected values (eqn ②)

→ The difference b/n the observed values and the Expected values is known as residual.

→ We want to minimise the residuals, since these difference may be positive in some cases. and negitive in others.

→ for our convienent to make the sum of squares of these residuals a minimum. this method is called Method of least squares.

$$S = \sum_{i=1}^{n} d_i^2$$

✕ Method of least Squares ✕

→ Fitting a straight line

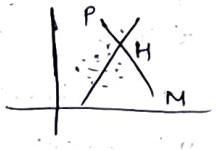
- Suppose the equation of the straight line is

$$y = a_0 + a_1 x \quad \text{—①}$$

- Suppose the Residual is $d_i = y_i - (a_0 + a_1 x_i)$

(y_i: observed value; $a_0 + a_1 x_i$: Predicted value)

- the eqn $y_i = f(x_i) = a_0 + a_1 x_i$ represents a family of straight lines for different values of arbitary constants.
- we have to determine a_0 and a_1 so that the eq ① is the line of best fit
- Let $p(x, y)$ be any point in the scattered diagram and draw the PM line which is ⊥lar to x-axis. meeting the line eq ① in H. and it's co-ordinates are $(x_i, a_0 + a_1 x_i)$



$P_H = PM - HM$

$= y_i - (a_0 - a_1 x_i) = d_i$ residual.

Suppose, the eqn of st line is

$$S = \sum_{i=1}^{n} d_i^2 = \sum_{i=1}^{n} (y_i - (a_0 + a_1 x_i)^2) \quad \text{—②}$$

for minima. we have

$$\frac{\partial s}{\partial a_0} = 0$$

$$= -2 \sum_{i=1}^{n} (y_i - (a_0 + a_1 x_i)) = 0$$

$$= \sum_{i=1}^{n} (y_i - (a_0 - a_1 x_i) = 0$$

$$= \sum_{i=1}^{n} (y_i - (a_0 + a_1 x_i)) = 0.$$

$$\sum_{i=1}^{n} y_i = \sum_{i=1}^{n} a_0 + a_1 \sum_{i=1}^{n} x_i$$

$$\sum_{i=1}^{n} y_i = n a_0 + a_1 \sum_{i=1}^{n} x_i \quad \text{—Ⓐ}$$

and $\frac{\partial s}{\partial a_1} = 0$

$$2 \sum_{i=1}^{n} [y_i - (a_0 + a_1 x_i)](-x_i) = 0$$

$$\sum_{i=1}^{n} x_i [y_i - (a_0 + a_1 x_i)] = 0$$

$$\sum_{i=1}^{n} x_i y_i = a_0 \sum_{i=1}^{n} x_i + a_1 \sum_{i=1}^{n} x_i^2 \quad \text{—Ⓑ}$$

- the eqns Ⓐ and Ⓑ are called normal eqns.
- The values $\sum_{i=1}^{n} x_i$, $\sum_{i=1}^{n} y_i$, $\sum_{i=1}^{n} x_i y_i$, $\sum_{i=1}^{n} x_i^2$ can be calculated from given data.
- After calculating these values, we can subsituting eq Ⓐ and eq Ⓑ, we get two equations. a_0 and a_1.
- After solving these eqns we get the values a_0 and a_1.

① A chemical company ~~whist~~ wishing to study the effects of the extraction time on the efficiently of an extraction operation, obtained the data shown in the following table.

Extraction time (minute) x	~~25~~	45	41	19	35	39	19	49	15	31
Extraction efficiency (y)	57	64	80	46	62	72	52	~~77~~	57	68

Fit a straight line to the given data by the method of least squares and use it to predict the exaction efficiency one. and can Expect the exaction time is 35 minutes.

Sol:- Let $y = a_0 + a_1 x$ —① be a st. line.

The normal eqns are

$$\Sigma y = n a_0 + a_1 \Sigma x$$

$$\Sigma xy = a_0 \Sigma x + a_1 \Sigma x^2$$

x	x^2	y	xy
27	729	57	1539
45	2025	64	2880
41	1681	80	3280
19	361	46	874
35	1225	62	2170
39	1521	72	2808
19	361	52	988
49	2401	77	3773
15	225	57	855
31	961	68	2108
$\Sigma x = 320$	$\Sigma x^2 = 11490$	$\Sigma y = 635$	$\Sigma xy = 21275$

$$\Sigma y = n a_0 + a_1 \Sigma x \text{ —②}$$

$$\Sigma xy = a_0 \Sigma x + a_1 \Sigma x^2 \text{ —③}$$

Normal eqns are $10 a_0 + 320 a_1 = 635$ —Ⓐ

Eq② $\Rightarrow 635 = 10 a_0 + a_1 (320)$

Eq③ $\Rightarrow 21275 = a_0 (320) + a_1 (11490)$

$$320 a_0 + 11490 a_1 = 21275 \text{ —Ⓑ}$$

solving eqn Ⓐ and eqn Ⓑ, we get

$$a_0 = 39.05$$

$$a_1 = 0.764$$

$\therefore y = a_0 + a_1 x$

$$\boxed{y = 39.05 + 0.764 x}$$

Now, when $x = 35$; $y = ?$

$$y = 39.05 + (0.764)(35)$$

$$y = 65.79$$

② Fit a st line for the following data.

x	1	2	3	4	5	6
y	14	33	40	63	76	85

Sol:- Let $y = a_0 + a_1 x$ —① be a st. line.

The normal eqns are

$$\Sigma y = n a_0 + a_1 \Sigma x$$

$$\Sigma xy = a_0 \Sigma x + a_1 \Sigma x^2$$

x	x^2	y	xy
1	1	14	14
2	4	33	66
3	9	40	120
4	16	63	252
5	25	76	380
6	36	85	510
$\Sigma x = 21$	$\Sigma x^2 = 91$	$\Sigma y = 311$	$\Sigma xy = 1342$

$$\Sigma y = n a_0 + a_1 \Sigma x \text{ —②}$$

$$\Sigma xy = a_0 \Sigma x + a_1 \Sigma x^2 \text{ —③}$$

Normal eqns are

Eqn ① ⇒ $674 = 6a_0 + a_1 21$

Eqn ② ⇒ $342 = 6a_0 + a_1 91$

$6a_0 + a_1 21 = 674$ — (A)

$6a_0 + a_1 91 = 342$ — (B)

$a_0 = \cancel{128.93}$ 1.12

$a_1 = \cancel{4.74}$ 14.49

$y = a_0 + a_1 x$

$y = 1.12 + 14.49x$

* Fitting a parabola (second degree curve)

$y = a_0 + a_1 x + a_2 x^2$

In a second degree parabola eqn.

⇒ The normal eqn's are

$$\sum_{i=1}^{n} y_i = n a_0 + a_1 \sum_{i=1}^{n} x_i + a_2 \sum_{i=1}^{n} x_i^2$$

$$\sum_{i=1}^{n} x_i y_i = a_0 \sum_{i=1}^{n} x_i + a_1 \sum_{i=1}^{n} x_i^2 + a_2 \sum_{i=1}^{n} x_i^3$$

$$\sum_{i=1}^{n} x_i^2 y_i = a_0 \sum_{i=1}^{n} x_i^2 + a_1 \sum_{i=1}^{n} x_i^3 + a_2 \sum_{i=1}^{n} x_i^4$$

① ⇒ Fit a second degree parabola using method of least Squares

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

Solt Let the curve be ~~$y = a_0 + a_1 x + a_2$~~

$y = a + bx + cx^2$

Normal eqn's are

$\Sigma y = na + b\Sigma x + c\Sigma x^2$

$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$

$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$

x	x^2	x^3	x^4	y	xy	x^2y
0	0	0	0	1	0	0
1	1	1	1	1.8	1.8	1.8
2	4	8	16	1.3	2.6	5.2
3	9	27	81	2.5	7.5	~~22.7~~
4	16	64	256	6.3	25.2	100.8
$\Sigma x = 10$	$\Sigma x^2 = 30$	$\Sigma x^3 = 100$	$\Sigma x^4 = 354$	$\Sigma y = 12.9$	$\Sigma xy = 37.1$	$\Sigma x^2 y = 128.5$

Normal eqn's ⇒ $\Sigma y = na + b\Sigma x + c\Sigma x^2$

$12.9 = 5a + 10b + 30c$

$5a + 10b + 30c = 12.9$ — (A)

$\Rightarrow \Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$

$37.1 = a10 + b30 + c100$

$10a + 30b + 100c = 37.1$ — (B)

$\Rightarrow \Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$

$128.5 = a30 + b100 + c354$

$30a + 100b + 354c = 128.5$ — (C)

Solve A, B, and C

$A = 1.42$, $B = -1.07$, $C = 0.55$

St. line $y = a + bx + cx^2 \Rightarrow 1.42 - 1.07x + 0.55x^2$

② Fit second degree parabola using method least sq.

x	1	2	3	4	5	6	7	8	9
y	2	6	7	8	10	11	11	10	9

Solt Let the curve be $y = a + bx + cx^2$

Normal eqn's are

$\Sigma y = na + b\Sigma x + c\Sigma x^2$

$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$

$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$

x	x^2	x^3	x^4	y	xy	x^2y
1	1	1	1	2	2	2
2	4	8	16	6	12	24
3	9	27	81	7	21	63
4	16	64	256	8	32	128
5	25	125	625	10	50	250
6	36	216	1296	11	66	396
7	49	343	2401	11	77	539
8	64	512	4096	10	80	640
9	81	729	6561	9	81	729
$\Sigma x = 45$	285	$\Sigma x^3 =$ 2025	$\Sigma x^4 =$ 15252	$\Sigma y = 74$	$\Sigma xy =$ 421	$\Sigma x^2y =$ 2771

normally eq are

$$\Sigma y = na + b\Sigma x + c\Sigma x^2$$

$$74 = 9a + 45b + 285c$$

$$9a + 45b + 285c = 74 \quad \text{—(1)}$$

① The profit of a certain company in the xth year of its life are given by

x:	1	2	3	4	5
y:	1250	1400	1650	1950	2300

Fit a parabolat

Sol: changing the orgin $u = x - 3$ ← Avg. $\Rightarrow \frac{15}{5} = 3$

$$v = \frac{y - 1650}{50}$$

here $u = -2, -1, 0, 1, 2$

then second degree parabola eqn can be written as.

$$v = a + bu + cu^2$$

normal Eqn's are

$$\Sigma v = na + b\Sigma u + c\Sigma u^2$$

$$\Sigma uv = a\Sigma u + b\Sigma u^2 + c\Sigma u^3$$

$$\Sigma u^2 v = a\Sigma u^2 + b\Sigma u^3 + c\Sigma u^4$$

x	y	u	v	u^2	u^3	u^4	uv	u^2v
1	1250	−2	−8	4	8	16	16	−32
2	1400	−1	−5	1	1	1	5	−5
3	1650	0	0	0	0	0	0	0
4	1950	1	6	1	1	1	6	6
5	2300	2	13	4	8	16	26	52
$\Sigma x =$ 15	$\Sigma y =$	$\Sigma u =$ 0	$\Sigma v =$ 6	$\Sigma u^2 =$ 10	$\Sigma u^3 =$ 0	$\Sigma u^4 =$ 34	$\Sigma uv =$ 53	$\Sigma u^2v =$ 21

(n = 5)

Subsitute these values in normal eqn's.

Normal eqn's are

$$6 = 5a + b(0) + c(10)$$

$$5a + 10c = 6 \quad \text{—(A)}$$

$$53 = a(0) + b(10) + c(0)$$

$$10b = 53 \quad \text{—(B)}$$

$21 = a(10) + b(0) + c(34)$

$10a + 34c = 21$ — (c).

Sub Solving these 3 equations.

$a = -0.08,\ b = 5.3\ ;\ c = 0.643$

$\therefore V = -0.086 + 5.3u + 0.643u^2$

$\frac{y - 1650}{50} = -0.86 + 5.3(x-3) + 0.643(x-3)$

$\frac{y}{50} = 33 - 0.086 + 5.3x - 15.9 + 0.643(x^2 - 6x + 9)$

$\boxed{y = 1140 + 72x + 32.15x^2}$

Curve linear Regression :-

we can also use the method of least squares to fit a polynomial and Exponential curves.

I) To fit the curve of the form $y = a.b^x$ — (1)

Take 'log' on b/s we get

$\log y = \log a + x \log b$.

$\boxed{Y = A_0 + A_1 X}$ — (2) where $Y = \log y$, $A_0 = \log a$, $A_1 = \log b$, $X = x$.

Normal eqn's are :-

$\Sigma y = nA_0 + A_1 \Sigma x$

$\Sigma xy = A_0 \Sigma x + A_1 \Sigma x^2$

1) Fit $y = ab^x$ by the method of least squares to following data.

x	0	1	2	3	4	5	6	7
y	10	21	35	59	92	200	400	610

Sol: let $y = a.b^x$ — (1)

Taking 'log' on b/s we get

$\log y = \log a + x \log b$

$Y = A_0 + A_1 X$ — (2) where $A_0 = \log a$, $A_1 = \log b$, $x = X$, $y = \log y$

There eqn to is a straight line eqn. Now we apply the method of least squares.

Normal eqn's are

$\Sigma y = nA_0 + A_1 \Sigma X$ — (3)

$\Sigma xy = A_0 \Sigma X + A_1 \Sigma X^2$ — (4)

x	y	X	$Y = \log y$	X^2	XY
0	10	0	1	0	0
1	21	1	1.322	1	1.322
2	35	2	1.5441	4	3.089
3	59	3	1.7708	9	5.31
4	92	4	1.963	16	7.84
5	200	5	2.301	25	11.50
6	400	6	2.602	36	15.6
7	610	7	2.785	49	19.46
		$\Sigma X = 28$	$Y = 15.28$	$\Sigma X^2 = 140$	$\Sigma XY = 64.296$

$15.2893 = 8A_0 + 28A_1$ — (I)

$64.1927 = 28A_0 + 140A_1$ — (II)

Solving eq (I) and eq (II) we get

$A_1 = 0.254$

$A_0 = 1.021$

Now, $A_0 = \log a$ | $A_1 = \log b$

$1.021 = \log a$ | $0.254 = \log b$

$a = 10^{1.021}$ | $b = 10^{0.254}$

$= 10.499$ | $b = 1.7959$

$\therefore y = ab^x$

$y = (10.499)(1.7959)^x$

II) To fit curve of type $y = a \cdot x^b$ — (1)

$\log y = \log a + b \log x$

$Y = A_0 + A_1 X$ — (2) where $Y = \log y$, $A_0 = \log a$, $\log x = X$, $b = A_1$

Normal eqn's are:

$\Sigma Y = n A_0 + A_1 \Sigma X$

$\Sigma XY = A_0 \Sigma X + A_1 \Sigma X^2$

III. To fit a curve of type $y = ae^{bx}$:

let $y = ae^{bx}$ — (1)

log ⇒ on b/s we get

$\log y = \log a + bx$

$\boxed{Y = A_0 + A_1 X}$ — (2) where $A_0 = \log_e a$, $A_1 = b$, $X = x$, $Y = \log_e y$

Normal equations are:

$\Sigma Y = n A_0 + A_1 \Sigma X$

$\Sigma XY = A_0 \Sigma X + A_1 \Sigma X^2$

CORRELATION

The relationship b/n the two variables such that the change in one variable results a +ve or -ve change in the other and also greater change in one variable results in a corresponding greater change in the another is called co-relation.

- For a change in a one variable, there is a corresponding change in the other variable. then the variables are said to be co-related.

a) The two variables deviated in the same direction, the co-relation is said to be direct or positive.

b) If the variables deviated the opp. direction the corcleation is said to be inverse or -ve.

c) The change in one variable corresponds to a proportional change in the other. then the correlation is said be perfect.

Karl pearson co-efficient of correlation:

- The numberical measure of linear relationship b/n the Random variables x, y is called coefficient of co-relation b/n the variables.
- we write as $r(X, Y)$ or r and it is defined as

$$\boxed{r = \frac{S_{xy}}{\sqrt{S_{xx}}\sqrt{S_{yy}}}} \text{ (or)}$$

$$\boxed{r = \frac{\text{Co-Variance}(X, Y)}{\sigma_x \sigma_y}}$$

$$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$S_{yy} = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

Covariance of $(X, Y) = E[(x - E(x))(y - E(y))]$

$$= \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

$$\sigma_x^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$\sigma_y^2 = \frac{1}{n} \sum_{i=1}^{n} (y_i - \bar{y})^2$$

$$\Rightarrow \sum_{i=1}^{n} (y_i - \bar{y})^2 = n \sigma_y^2$$

NOTE:

① The coefficient of corelation lies b/n -1 and 1.

② if $r = 0$, the variables are not co-rrelation (or) There is no correlation b/n the variably X and Y.

③ If $r = 1$, we said that posi The Variables are +rely and perfectly co-related.

④ If $r = -1$, we say that The variables are -vely and perfectly co-related.

⑤ If $-1 < r < 0$ then the Variables are -rely, perfectly Corelated.

⑥ If $0 < r < 1$, then The variables are +rely, partially co-related.

① The marks obtained by the 10 students in mathematics and statistics are given bdow. Find the coeffient of corelation b/n The 2 subs.

Marks in Maths X	75	30	60	80	53	35	15	40	38	48
Marks in stats Y	85	45	54	91	58	63	35	43	45	44

Solr

$$\bar{x} = \frac{1}{n} \Sigma x$$

$$= \frac{1}{10} (75 + 30 + 60 + \dots + 38 + 48)$$

$$= \frac{474}{10} = 47.4$$

$$\bar{y} = \frac{1}{n} \Sigma y$$

$$= \frac{1}{10} [85 + 45 + 54 + 91 + 58 + \dots + 44)$$

$$= \frac{563}{10} = 56.3$$

x_i	y_i	$x_i = x_i - \bar{x}$	$y_i = y_i - \bar{y}$	$x_i y_i$	x^2 / $x x_1$	y^2 / $y_i y_i$
75	85	27.6	28.7	792.12	761.76	823.69
30	45	-17.4	-11.3	196.62	302.76	127.69
60	54	12.6	-2.3	-28.98	158.76	5.29
80	91	32.3	34.7	1120.81	1043.29	1204.09
53	58	-47.4	1.7	-80.58	2246.76	2.89
35	63	-12.4	6.7	-83.08	153.76	44.89
15	35	-32.4	-2.13	690.12	1049.76	45.369
40	43	-7.4	-13.3	98.42	54.76	176.89
38	45	-9.4	-11.3			
48	44					

$$\Sigma xy = \sum_{i=1} (x_i - \bar{x})(y_i - \bar{y})$$

$$= \sum_{i=1} x_i y_i$$

$$\Sigma xy = 2911.56$$

and $$S_{xx} = \sum_{i=1} (x_i - \bar{x})^2$$

$$= \sum_{i=1} x_i^2$$

$$S_{xx} = 3664.40.$$

$$S_{yy} = \sum_{i=1} (y_i - \bar{y}) = \sum_{i=1}^{n} y_i^2$$

$$S_{yy} = 3118.10$$

Coefficient of corr. $r = \dfrac{2911.56}{\sqrt{3664.40}\sqrt{3118.10}}$

$$\boxed{r = 0.86}$$

The relation b/n x & y is +ve & partially corelated.